

Nextcloud Troubleshooting & Optimization Log

Server: kitcld.kayson-ir.com

Platform: Ubuntu + Apache + PHP 8.3 + Nextcloud Hub 34.0.2

Period: LDAP integration through current optimization work

1. LDAP / Active Directory Integration

Purpose

Allow users to authenticate against Microsoft Active Directory.

Actions Taken

- Configured LDAP integration.
- Verified LDAP connection.
- Investigated inactive LDAP configuration warning.

Result

- LDAP authentication working.
- Inactive LDAP configurations should be removed if not used.

Check LDAP Configurations

```
sudo -u www-data php /var/www/nextcloud/occ ldap:show-config
```

Delete Unused Configuration

```
sudo -u www-data php /var/www/nextcloud/occ ldap:delete-config s01
```

2. OpenAI Assistant Integration Failure

Purpose

Enable AI Assistant features.

Problem

Logs showed:

429 Too Many Requests

You exceeded your current quota

Tasks failed:

core:text2text:chat

core:text2text:translate

Assistant UI showed:

This chat response is taking longer to start generating than expected

Root Cause

OpenAI API quota exhausted.

Resolution Options

Option A - Fix Billing

Add credits to OpenAI account.

Option B - Disable AI Apps

List AI apps:

```
sudo -u www-data php /var/www/nextcloud/occ app:list
```

Disable:

```
sudo -u www-data php /var/www/nextcloud/occ app:disable assistant
```

```
sudo -u www-data php /var/www/nextcloud/occ app:disable integration_openai
```

3. Exchange EWS Connector Installation

Purpose

Connect Microsoft Exchange calendars and mail.

Problem

Installation failed:

The library soap is not available

Resolution

Install PHP SOAP module:

```
sudo apt update
```

```
sudo apt install php8.3-soap
```

Enable module:

```
sudo phpenmod soap
```

Restart Apache:

```
sudo systemctl restart apache2
```

Later Decision

EWS integration was removed.

4. PHP Service Restart Investigation

Purpose

Restart PHP after configuration changes.

Finding

Server runs Apache mod_php.

No PHP-FPM service exists.

Command failed:

```
systemctl restart php8.2-fpm
```

Correct Restart

```
sudo systemctl restart apache2
```

Verify:

```
php -v
```

Output:

PHP 8.3.6

5. Redis Memory Cache Configuration

Purpose

Improve performance and file locking.

Warning Fixed

No memory cache configured

Transactional File Locking uses database

Installation

```
sudo apt install redis-server php-redis
```

Enable Redis:

```
sudo systemctl enable redis-server
```

```
sudo systemctl start redis-server
```

Config.php Changes

Added INSIDE the \$CONFIG array:

```
'memcache.local' => '\OC\Memcache\APCu',
```

```
'memcache.distributed' => '\OC\Memcache\Redis',
```

```
'memcache.locking' => '\OC\Memcache\Redis',
```

```
'redis' => [  
    'host' => 'localhost',  
    'port' => 6379,  
],
```

Mistake Encountered

Configuration added outside \$CONFIG.

Error:

ParseError: syntax error, unexpected token =>

Fix

Moved Redis configuration inside \$CONFIG array.

Result:

- Redis enabled.
- File locking warning resolved.
- Memory cache warning resolved.

6. Missing Database Indexes

Purpose

Improve database query performance.

Command

```
sudo -u www-data php /var/www/nextcloud/occ db:add-missing-indices
```

Result

Missing index warnings removed.

Example:

```
tables_row_sleeves_t_id
```

7. Mimetype Migration

Purpose

Improve handling of newer file types.

Command

```
sudo -u www-data php /var/www/nextcloud/occ maintenance:repair --include-expensive
```

Result

Mimetype migration warnings removed.

8. Cron Job Verification

Purpose

Ensure background jobs execute correctly.

Verification

```
systemctl status cron
```

Observed:

```
php -f /var/www/nextcloud/cron.php
```

Running every 5 minutes.

Result

Cron functioning normally.

9. Brute Force Throttling Warning

Warning

Remote address 172.16.20.97 is brute-force throttled

Cause

Repeated failed logins.

Could also indicate reverse proxy configuration issue.

Clear Attempts

```
sudo -u www-data php /var/www/nextcloud/occ security:bruteforce:reset 172.16.20.97
```

Additional Check

Verify reverse proxy settings:

```
'trusted_proxies' => [
```

```
    'PROXY_IP'
```

```
],
```

```
'forwarded_for_headers' => [
```

```
    'HTTP_X_FORWARDED_FOR'
```

],

10. OPcache Optimization

Warning

opcache.interned_strings_buffer nearly full

Fix

Edit:

```
sudo nano /etc/php/8.3/apache2/php.ini
```

Find:

```
opcache.interned_strings_buffer=8
```

Change:

```
opcache.interned_strings_buffer=16
```

Optional:

```
opcache.memory_consumption=256
```

Restart:

```
sudo systemctl restart apache2
```

11. Maintenance Window

Purpose

Run heavy jobs outside business hours.

Config.php

Add:

```
'maintenance_window_start' => 2,
```

Meaning:

02:00 AM

12. Default Phone Region

Warning

No default phone region set

Fix

Add:

'default_phone_region' => 'IR',

Replace with your country code if different.

13. HSTS Security Header

Purpose

Improve HTTPS security.

Apache SSL Virtual Host

Add:

Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"

Enable headers:

```
sudo a2enmod headers
```

Restart:

```
sudo systemctl restart apache2
```

14. Email Configuration

Warning

Email could not be sent

Investigation Steps

Check Nextcloud settings:

```
sudo -u www-data php /var/www/nextcloud/occ config:list system | grep -A20 mail
```

Check logs:

```
sudo tail -f /var/www/nextcloud/data/nextcloud.log
```

Check mail server:

```
sudo tail -50 /var/log/mail.log
```

Microsoft 365 Recommended Settings

SMTP Server:

smtp.office365.com

Port:

587

Encryption:

STARTTLS

Authentication:

Enabled

Username:

user@domain.com

15. Whiteboard WebSocket Warning

Purpose

Enable real-time collaborative whiteboards.

Status

Not required unless using Whiteboard collaboration.

Can be ignored safely.

16. High Performance Backend (Talk)

Purpose

Scale video calls beyond 2–3 participants.

Current Status

Not configured.

Impact

Only affects Nextcloud Talk meetings.

No impact on files or general performance.

17. Client Push Warning

Purpose

Instant notifications.

Impact

Without it:

- Notifications delayed.
- Nextcloud still works.

Optional Future Installation

Install:

- notify_push

Only recommended for larger deployments.

18. AppAPI Deploy Daemon

Purpose

Required only for External Apps (Ex-Apps).

Impact

No impact unless using Ex-Apps.

Can be ignored.

19. Two-Factor Authentication

Purpose

Improve account security.

Status

Available but not enforced.

Recommendation

Enable for administrators.

20. Configuration Server ID

Purpose

Used in multi-server environments.

Impact

Single server installations can ignore.

Optional:

```
'config_is_read_only' => false,
```

```
'server_id' => 'kitcld01',
```

Current High-Priority Remaining Tasks

1. Fix SMTP email sending.
2. Increase OPcache interned_strings_buffer to 16.
3. Set maintenance_window_start.
4. Clear remaining log errors.

5. Configure HSTS header.
6. Investigate Task Processing failures (leftover AI tasks).
7. Reset brute-force throttle warning.
8. Configure High Performance Backend only if using Nextcloud Talk.
9. Configure Client Push only if instant notifications are needed.

Performance Improvements Completed

- ✓ Server upgraded to 16 GB RAM
- ✓ Server upgraded to 6 CPU cores
- ✓ Redis configured
- ✓ Transactional file locking optimized
- ✓ Missing database indices fixed
- ✓ Mimetype migrations fixed
- ✓ Cron verified operational
- ✓ PHP configuration reviewed
- ✓ AI-related failures identified